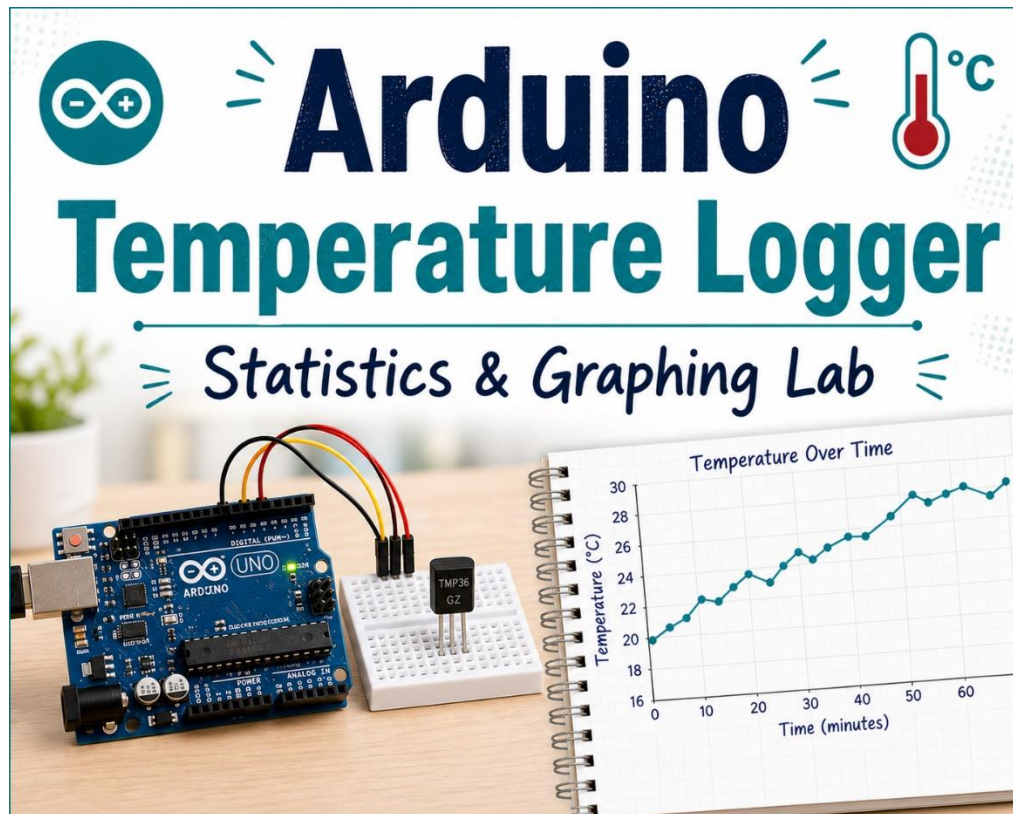


FREE SAMPLE PREVIEW

Arduino Temperature Logger

Statistics & Graphing Lab

A Real-Data Statistics Unit Using Arduino Sensors



GRADES
6 - 8

DURATION
3-4 Days

SUBJECT
Math + STEM

PAGES
18+ Pages

No Hardware? No Problem.
Tinkercad Option Included

Print + Digital Ready
Works in Google Classroom

Standards-Aligned
CCSS + NGSS + ISTE

ABOUT THIS RESOURCE

This lab positions the Arduino as a math tool — not just a coding device. Students collect real temperature data from two environments, then apply key statistics to analyze and compare their results. Every calculation is anchored to data they personally measured, making abstract statistics immediately intuitive and relevant.

WHAT THIS RESOURCE TEACHES

STATISTICS SKILLS

Mean Students calculate the average of 15 real temperature readings	Median Find the middle value using an ordered data set
Mode Identify the most repeated reading (or determine 'no mode')	Range Calculate variability across both sensor locations

DATA & GRAPHING SKILLS

Line Graphing Plot two datasets on a dual-line graph with proper labeling	Trend Analysis Interpret patterns and temperature changes over time
Two-Dataset Comparison Compare indoor vs. near-window environments with statistics	Sample Size & Reliability Explore how more data increases statistical confidence

STEM INTEGRATION

Real Data Collection Arduino TMP36 sensor or Tinkercad simulation	Arduino Code Modification Adjust delay timing to change sampling intervals
Serial Monitor Use Read and record live sensor output	Tinkercad Alternative Full lab available with zero hardware on Chromebooks/tablets

CCSS 6.SP.B.5

Summarize numerical data using measures of center and variability

NGSS Practice 3

Planning & carrying out investigations with real data

ISTE Standard 5

Computational thinker — uses tech to analyze & solve

WHAT'S INCLUDED IN THE FULL RESOURCE

TEACHER GUIDE

Teacher Guide (4 pages)

- ✓ Resource overview & learning objectives
- ✓ Standards alignment table (CCSS + NGSS + ISTE)
- ✓ 3-4 day pacing guide with minute-by-minute outline
- ✓ Differentiation tips — support strategies & enrichment
- ✓ Common misconceptions to address proactively
- ✓ Assessment suggestions (formative, exit ticket, summative)

STUDENT PACKET

Student Lab Packet (8+ pages)

- ✓ Part A: Arduino wiring reference + code activity
- ✓ Starter code — copy-ready for Arduino IDE
- ✓ Code Questions (3 critical thinking questions)
- ✓ Part B: Dual data tables — 2 locations x 15 readings
- ✓ Mean, median, mode & range calculations with work space
- ✓ Comparison table with all 4 statistics side-by-side

TINKERCAD GUIDE

Tinkercad Simulation Guide

- ✓ Step-by-step circuit setup screenshots
- ✓ How to simulate 3 different temperature environments
- ✓ Slider settings for indoors, near window, outdoor
- ✓ Tips for managing a virtual lab class successfully
- ✓ Compatible with Chromebooks, tablets & any browser
- ✓ Zero cost — no hardware or software purchases required

ANSWER KEY + RUBRIC

Answer Key + Scoring Rubric

- ✓ Complete Part A answers (all 3 code questions)
- ✓ Sample dataset with full worked solutions
- ✓ Model comparison table with realistic sample values
- ✓ Part C analysis model answers (4 questions + extension)
- ✓ 4-criterion scoring rubric (Mean, Median, Mode, Range)
- ✓ Teacher-only — not included in student packet

PART A — Set Up Your Temperature Sensor

Wire your Arduino and verify your sensor is working

Name: _____

Date: _____

Period: _____

GOAL

Wire your Arduino and TMP36 sensor, upload the starter code, and verify it works by reading temperatures in the Serial Monitor.

Wiring Reference Table

TMP36 Pin	Connects To	Wire Color
Left — V+ (power)	5V pin on Arduino	Red
Middle — Vout (signal)	A0 (Analog pin 0) on Arduino	Yellow / Orange
Right — GND (ground)	GND pin on Arduino	Black

Starter Code

Copy this code exactly into the Arduino IDE, then click Upload. Open the Serial Monitor to see readings.

```
// Math-O-Tronics | Temperature Logger Starter Code
void setup() {
  Serial.begin(9600);  // Start Serial Monitor
}
void loop() {
  int rawValue = analogRead(A0);
  float voltage = rawValue * (5.0 / 1023.0);
  float tempC = (voltage - 0.5) * 100.0;
  float tempF = (tempC * 9.0 / 5.0) + 32.0;
  Serial.print("Temp (F): ");
  Serial.println(tempF);
  delay(5000); // <-- STUDENT TASK: What does this control?
}
```

Code Questions

Q1

The line `delay(5000)` pauses the program. The number is in milliseconds (1000 ms = 1 second). How many seconds does it wait between readings?

Answer: _____

Q2

[Full question available in complete resource]

Q3

[Full question available in complete resource]

SAMPLE STUDENT PAGES | PART B — DATA TABLES

GOAL
Collect 15 temperature readings from two different locations. Then calculate mean, median, mode, and range for each dataset and compare.

Step 1 — Location 1 Data Table

Open the Serial Monitor. Wait for each reading, then record it below.

#	Reading (°F)	#	Reading (°F)	#	Reading (°F)	Location Name
1		6		11		Name:
2		7		12		
3		8		13		
4		9		14		
5		10		15		

Step 2 — Calculate Statistics for Location 1

MEAN (AVERAGE)

Add all 15 temperatures, then divide by 15.
Work space:

Sum = _____ Mean = _____ °F

MEDIAN (MIDDLE VALUE)

Write 15 readings smallest to largest. Circle the 8th value.

_____ [8th] _____

Median = _____

MODE

[Full statistics section in complete resource]

RANGE

[Full statistics section in complete resource]

WHY TEACHERS LOVE THIS RESOURCE

REAL DATA

Real-World Data That Matters

Students collect actual temperature readings — not made-up textbook numbers. Statistics become personally relevant and immediately meaningful when every data point comes from the student's own sensor.

No Hardware? Absolutely No Problem.

The full Tinkercad simulation alternative lets any class run this lab on Chromebooks or tablets — zero cost, zero hardware. Perfect for schools without maker space budgets or device limitations.

STANDARDS

Triple Standards Alignment — In One Lab

Hits CCSS 6.SP.B.5, NGSS Practice 3, and ISTE Standard 5 in one cohesive activity. Saves you hours of planning and satisfies multiple curriculum requirements simultaneously.

Differentiation Built Right In

Support strategies (pre-filled data, calculator-use guidance, pair strategies) and enrichment options (MAD, dot plots, formal lab report) are included in the teacher guide — no extra prep needed.

SUB-FRIENDLY

Sub-Friendly Step-by-Step Design

Clear, numbered student directions mean a substitute teacher can facilitate Parts B & C without any prior STEM experience. Great for planned absences and PLCs.

Print & Digital — Works Both Ways

Works beautifully as a printed workbook or in Google Classroom. No reformatting required for hybrid or remote learning. One purchase, two delivery modes.

***"My students were genuinely excited about statistics for the first time.
They kept asking to collect more data."***

— 7th Grade Math Teacher, Texas

Ready to Make Statistics Meaningful?

Turn abstract numbers into real discoveries — with data your students actually collected.

GET THE FULL RESOURCE TODAY

Full Resource Includes:

- ✓ Teacher Guide (4 pages)
- ✓ Student Lab Packet (8+ pages)
- ✓ Tinkercad Simulation Guide
- ✓ Complete Answer Key (Teacher Only)
- ✓ 4-Criterion Scoring Rubric
- ✓ Standards Alignment Table
- ✓ 18+ Total Pages

Perfect For:

- ✓ Grades 6, 7, and 8 Math classes
- ✓ STEM, Makerspace & elective courses
- ✓ Project-based learning units
- ✓ Schools with or without Arduinos
- ✓ Chromebook / tablet classrooms
- ✓ Substitute teacher days
- ✓ Remote or hybrid learning

MATH-O-TRONICS

STEM + Math Integration Resources | Grades 6-8
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PRINT + DIGITAL
Works in Google Classroom

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